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Effect of V_2O_5 addition on the crystallisation of glasses belonging to the $CaO-ZrO_2-SiO_2$ system

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Abstract

The crystallisation of $CaO-ZrO_2-SiO_2$ glasses doped with V_2O_5 (0.1–5 mol%) has been investigated in terms of microstructure and thermal parameters. Results indicate that crystallisation is predominantly controlled by a surface nucleation mechanism, even though a partial bulk nucleation has been encountered in compositions containing more than 2 mol% of doping oxide. As detected from differential thermal analysis curves, glass transition temperature and crystallisation temperature, are strongly dependent upon V_2O_5 content varying from 0.0 to 2.0 mol%, while the crystallisation activation energy values decrease with a parabolic trend from B-glass (0.0 mol% V_2O_5 content, 495 ± 7) to V-0.7 (0.7 mol% V_2O_5 content, 420 ± 6) composition, increasing again to 442 ± 5 kJ/mol K with higher amount of V_2O_5 . The microstructure of the glass-ceramic materials clearly showed a marked dependence upon the amount of V_2O_5 , also due to the presence of phase separation for content higher than 0.7 mol%. Wollastonite, $CaO \cdot SiO_2$, and a calcia–zirconia–silicate, $2CaO \cdot 4SiO_2 \cdot ZrO_2$, are the main crystalline phases whose ratio slightly varies with vanadium oxide content. The glass ceramics obtained from the studied materials are greenish and bluish coloured, so it is possible to use the studied glasses as coloured frits for tile glazes.

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1. Introduction

Research and development have covered new applications for traditional glass-ceramic materials, but relevant progress has also been made in the area of new formulation and process development, so that a variety of glasses can be manufactured into useful forms.

A recent type of a silicate glass belonging to $CaO-ZrO_2-SiO_2$ system has been developed for

the use in the industrial tile as a low-expansion glass-ceramic glaze or structural monolithic material [1].

The ternary phase diagram is known since the '50 [2], but glassy materials of compositions within the diagram have never been studied in a systematic way in spite of the fact that the presence of ZrO_2 leads to chemical resistance improvements in glasses as well as an amelioration of their optical and mechanical properties.

This paper intends to complete a previous study [3], where the authors investigated the colouring capability of vanadium when added as oxide to the above mentioned ternary glass system, by

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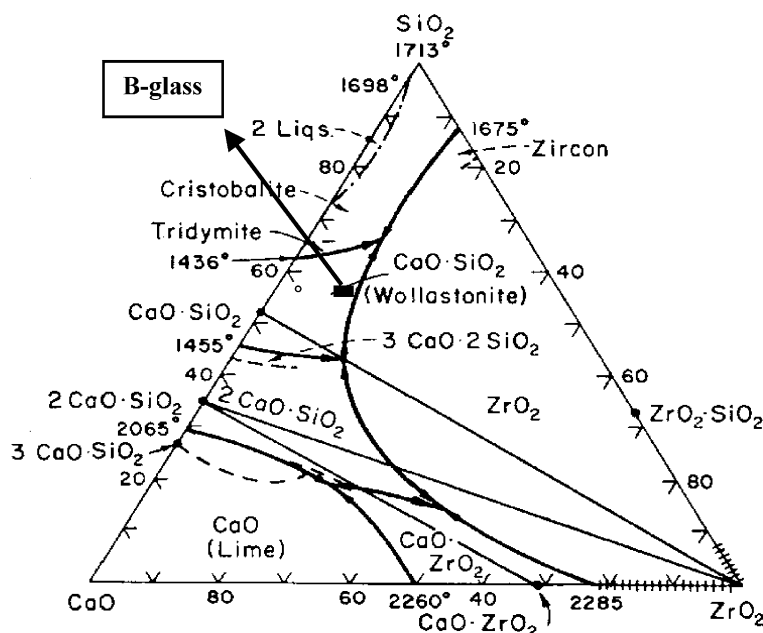


Fig. 1. B-glass composition in the CaO–ZrO₂–SiO₂ phase diagram.

deepening the investigation in the sense of the crystallisation tendency of the amorphous material. The choice of the undoped, hence colourless, glass composition is reported elsewhere and refers primarily to the meltability of the oxidic raw material in a conventional melting furnace [1]. The transition metal oxide, V₂O₅, has been chosen for the present work since it is the first transition colouring element in the periodic table. The effects of V₂O₅ on the glass thermal behaviour in term of crystallisation kinetics and microstructure have been studied for progressive additions of vanadium salts to the batch of the undoped ternary composition maintaining unvaried the CaO:ZrO₂:SiO₂ ratio.

A secondary aim of this work is to explore both the eventual nucleating effect of such an oxide and/or the phase separation process in terms of the current glass structure theories.

2. Experimental procedures

The undoped glass composition studied in this work, hereafter indicated as B-glass, belongs to the

wollastonite stability field of the CaO–ZrO₂–SiO₂ phase diagram (Fig. 1). The B-glass was added of different amounts of V₂O₅, equal to 0.1, 0.3, 0.5, 0.7, 1.0, 2.0 and 5.0 mol%, before melting (Table 1). All compositions were prepared by melting reagent grade raw materials: calcium carbonate (CaCO₃, 100%), zirconium silicate (ZrSiO₄, 100%), quartz (SiO₂, 99.9%) and ammonium metavanadate (NH₄VO₃, 100%). Batches, previously dry milled, were melted in an electrically heated furnace within Pt crucibles, following the same heating cycle: from 20 to 500 °C at 5 °C/min, from 500 to 1500 °C at 10 °C/min and finally 30 min of soaking time at the

Table 1
Composition of glasses (mol%)

Sample	SiO ₂	CaO	ZrO ₂	V ₂ O ₅
B-glass	57.17	36.75	6.08	0.00
V-0.1	57.11	36.71	6.08	0.10
V-0.3	57.00	36.64	6.06	0.30
V-0.5	56.88	36.57	6.05	0.50
V-0.7	56.77	36.49	6.04	0.70
V-1.0	56.60	36.39	6.02	0.99
V-2.0	56.05	36.03	5.96	1.96
V-5.0	54.45	35.00	5.79	4.76

maximum temperature of 1500 °C. The melts were quenched either into a graphite mould or in water in order to obtain bars or frits, respectively.

X-ray diffraction (XRD) and examination by scanning electron microscopy (SEM) showed no evidence of unmelted or crystalline particles in as-quenched glass except for the V-5.0 sample.

The quenched glass was ground and screened to four different particles size: <20, 30–40, 75–150, >400 µm (hereafter indicated as 20, 35, 100, 400), which were stored in an desiccator until used for DTA measurements.

The effect of particle size on the glass DTA peak profile was investigated by heating a constant weight of 30 mg of glass particles of different size in the DTA apparatus at 10 °C/min from room temperature until crystallisation was complete. Platinum containers were used for both the glass and the reference samples in air static atmosphere. The DTA equipment was calibrated periodically using Na₂SO₄ as a standard at the same conditions used for the samples measurements. The data were recorded and analysed in a computer interfaced with the DTA equipment. Samples used for DTA were as-quenched glass and they did not receive any pre-nucleation treatment.

To determine the kinetic parameters for the crystallisation, the temperature T_p , corresponding to the maximum of the crystallisation peak of each glass was determined at different heating rates, ϕ , (5, 10, 20 °C/min) for 20-µm-particle size glass samples. Such a small particle dimension assures the preference toward surface nucleation [4]. The error in the determination of the maximum position is due primarily to the accuracy of the DTA, which is ± 2 °C.

Data were plotted by using the Kissinger equation adapted for surface nucleation expressed as [5]

$$\ln(\phi/T_p^2) = -E/RT_p + \text{constant}, \quad (1)$$

where T_p is the temperature of the crystallisation peak, ϕ the heating rate, E the activation energy barrier for the crystallisation and R the gas constant. According to the Kissinger equation, a plot of $\ln(\phi/T_p^2)$ vs. $1/T_p$ should be linear, and the slope is $-E/R$.

To analyse the formation of crystalline phases during the heating process, in situ high-tempera-

ture XRD was performed on powdered glass. Powders were heated at 10 °C/min up to 1100 °C. At this temperature patterns were collected using a powder diffractometer with a Ni-filtered Cu K α radiation in the 20–40° 2 θ range, step size 0.02°, 3 s time step.

In order to obtain glass-ceramic materials, small plates (1 mm thick) of annealed glass were cut from the bars using a low-speed diamond saw with an emulsion of oil and water as lubricant, and then they were polished down to <0.5 µm alumina paste. All samples were inserted into the muffle kiln at room temperature, and successively heated at 10 °C min⁻¹ at different temperatures: 800, 850, 900, 950 and 1000 °C for 60 min. Specimens were then removed at the desired temperature and time and placed on refractory brick up to room temperature. None of the samples fractured during cooling.

To detect and identify the crystalline phases formed during the heat treatments, XRD was performed on finely ground glass-ceramic specimens. Patterns were collected using a powder diffractometer with a Ni-filtered Cu K α radiation in the 10–50° 2 θ range, step size 0.02°, 3 s time step.

Information concerning the mechanism of crystallisation and the microstructure of the glass-ceramic materials were obtained by a SEM, on superficially polished gold-coated specimens. Energy dispersion X-ray spectroscopy, EDS, was used to identify the chemical composition of the different phases in the samples.

The microstructure of V-5.0 glassy sample has been carried out by transmission electron microscopy (TEM), using an electron microscope operating at 200 kV. Thin foils transparent to the electrons have been prepared using standard techniques. It was necessary to deposit a thin layer of carbon onto the specimen surface, with the aim to increase the thermal and the electric conductivities of thin foils.

3. Results

The undoped colourless B-glass, acquires coloration from light green to dark brown with increasing the V₂O₅ content. All the glasses containing

V_2O_5 up to 2 mol% are transparent and the colour obtained is stable during cooling; on the contrary the quenched materials of V-5.0 composition are opaque and dark in colour.

XRD patterns of all samples show the characteristic broad band of the amorphous state even though SEM observations on polished samples evidence for V-5.0 composition a phase separation (Fig. 2). A more detailed observation of the phenomenon indicates the formation of a fine dispersion of spherical particles homogeneously distributed in the matrix (less than 1 μm in diameter) which appear in dark colour at the secondary electron (SE) detector. The EDS analysis of these particles shows the prevalence of the lightest element, silicon, while the surrounding glassy phase contains also the other elements.

Fig. 3(a) and (b) compare the typical selected area diffraction patterns (SAD) arising from the interiors of the particles and from the matrix, respectively. In both cases, the absence of crystallinity is evidenced.

From this point on, the V-5.0 glass composition will not be considered in the following thermal characterisation since its inhomogeneity makes further observation too complicate.

DTA thermograms of the studied compositions collected on the powders at a heating rate of 10 $^{\circ}\text{C}/$

min present very similar features: (a) an endothermic peak, T_g , at about 800 $^{\circ}\text{C}$, corresponding to the glass transition temperature; (b) an exothermic event with a maximum in the range 900–1000 $^{\circ}\text{C}$, indicating the crystallisation temperature, T_p . Moreover it is possible to consider that the crystallisation peak of the B-glass is a little broader than those recorded for samples containing the dopant oxide (Fig. 4).

Glass transition temperature, T_g and crystallisation temperature T_p are remarkably influenced by V_2O_5 content in the glasses, even though a non-linear behaviour is recorded (Figs. 5 and 6). The decrease in the maximum height of the DTA crystallisation peak (δT)_p measured at 10 $^{\circ}\text{C}/\text{min}$ for as-quenched glasses, with the increasing of the glass particle size, is shown in Fig. 7.

The temperature corresponding to the maximum of the DTA crystallisation peak, T_p , increases with increasing particle size, in a different way for V-0.1 and V-2.0 glasses (Fig. 8): as a matter of fact V-0.1 shows a more pronounced increasing with respect to V-2.0 sample.

The Kissinger plots for all samples heated at 5, 10 and 20 $^{\circ}\text{C}/\text{min}$ are shown in Figs. 9 and 10. The E values determined from the slope of these plots are given in Table 2. Fig. 9 shows the Kissinger plots for the B-glass, V-0.1, V-0.3 samples while

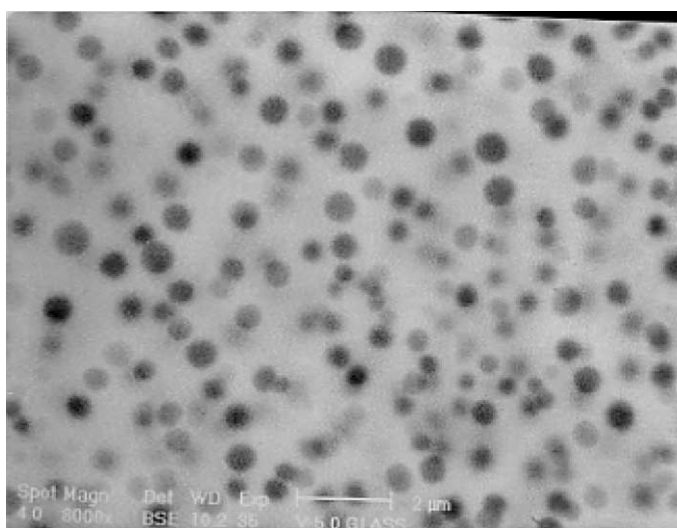


Fig. 2. SEM micrograph of V-5.0 glass shows the observed liquid–liquid phase separation.

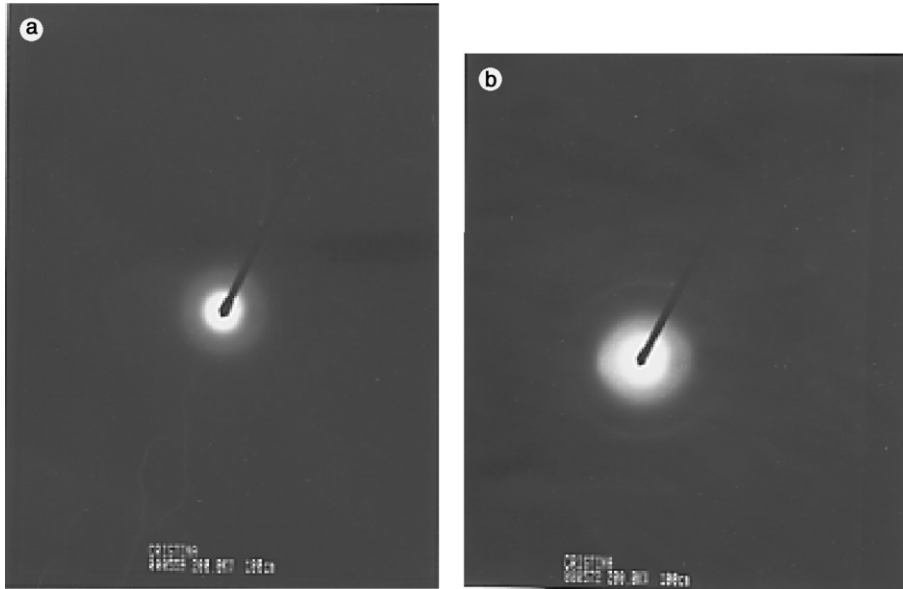


Fig. 3. Diffraction patterns (SAD) of V-5.0 glass: (a) interior of particles and (b) matrix.

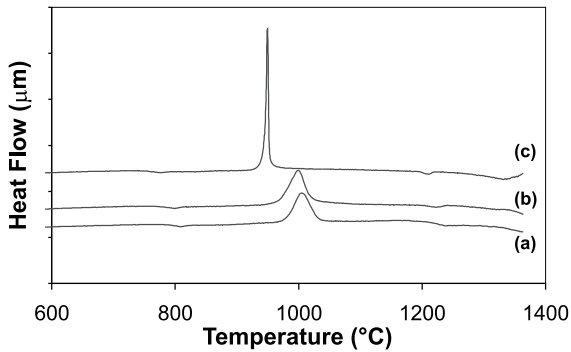


Fig. 4. DTA thermograms of the glasses (grain size < 20 μm): (a) base-glass, (b) V-0.5 and (c) V-1.0 samples.

Fig. 10 shows the V-0.5, V-0.7, V-1.0, and V-2.0 ones. Plots in both cases are almost parallel and quite close each other, and consequently, E , for the first set of samples reported in the first plot (Fig. 9) is the same within the experimental error (490 ± 5 kJ), while for the second one the activation energy ranges from 420 ± 6 to 442 ± 5 kJ.

No traces of crystalline phases were recorded on XRD patterns, for temperatures lower than 850 $^{\circ}\text{C}$ for all compositions; only V-5.0 sample shows a weak crystallisation. This technique gives

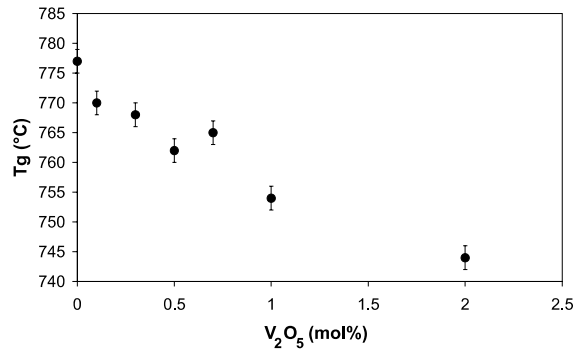


Fig. 5. Glass transition temperature, T_g , vs. V_2O_5 glass content.

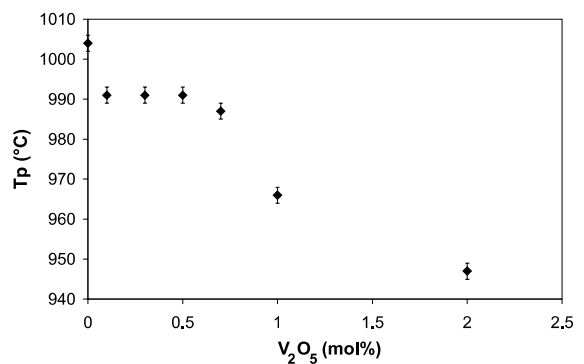


Fig. 6. Crystallisation temperature, T_p , vs. V_2O_5 glass content.

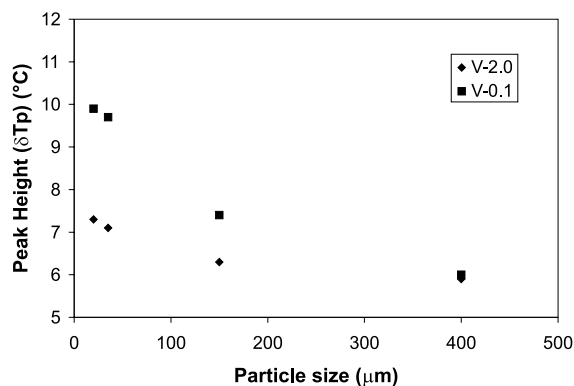


Fig. 7. The maximum height of the crystallisation peak vs. particle size.

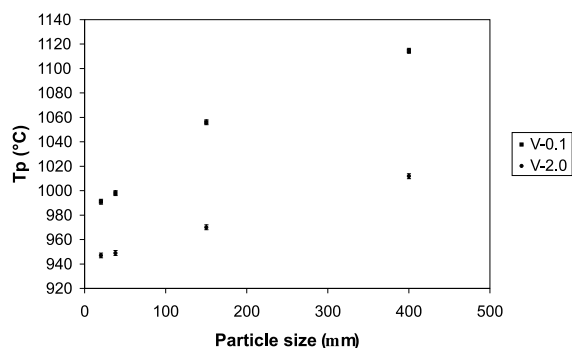


Fig. 8. Crystallisation temperature, T_p , vs. particle size of V-0.1 and V-2.0 glasses.

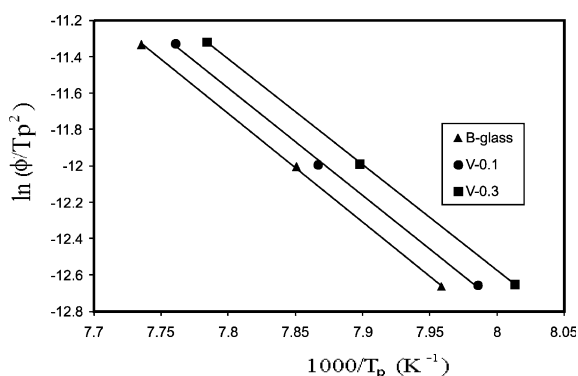


Fig. 9. Kissinger plot of data obtained from DTA curves of base, V-0.1, V-0.3, V-0.5 glasses.

moreover a qualitative indication about the amorphous phase present in the samples. The relative

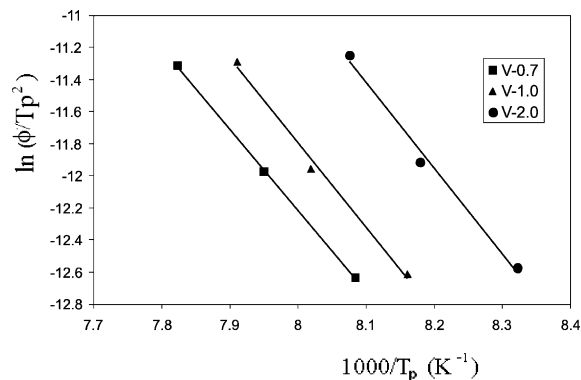


Fig. 10. Kissinger plot of data obtained from DTA curves V-0.7, V-1.0, V-2.0 glasses.

Table 2
Crystallisation activation energy of glasses

Glasses	E_a (kJ/mol K)
B-glass	495 ± 7
V-0.1	491 ± 6
V-0.3	482 ± 4
V-0.5	423 ± 3
V-0.7	420 ± 6
V-1.0	438 ± 6
V-2.0	442 ± 5

abundance of these crystalline and glassy phases, varies depending both on the compositions of the parent glass and the devitrification temperature. A summary of crystalline evolution for the all compositions is given in Table 3. At the highest temperature, 1000 °C, all glasses devitrify.

The first appearance of a massive formation of crystalline phases indicates the presence of a calcium–silicate phase, wollastonite (ICCD files nos. 10–489, 27–88), for B-glass and V-0.1 compositions, which is the main crystalline phase up to V-0.7. From V-0.3 composition another crystalline phase, a calcium zirconium silicate, $2\text{CaO} \cdot 4\text{SiO}_2 \cdot \text{ZrO}_2$ (ICCD file nos. 39–194) appears as a secondary phase, which becomes the main crystalline phase at V_2O_5 content higher than 0.7 mol%. From V-1.0 to V-5.0 compositions, wollastonite crystals progressively transform into pseudowollastonite (ICCD file nos. 31–300); moreover a minor phase is present in V-2.0 and V-5.0 glass

Table 3
Main crystalline phases present into glass-ceramics materials

Glass	800	850	900	950	1000
B-Glass	A	A	A	A	W
V-0.1	A	A	A	A W (traces)	W
V-0.3	A	A	A	Glass phase W, CZS	W, CZS
V-0.5	A	A	A	Glass phase W, CZS	CZS, W
V-0.7	A	A	Glassy phase W, CZS	W, CZS	CZS, W
V-1.0	A	A	Glassy phase W, CZS	CZS, W	CZS, W
V-2.0	A	A	CZS, PW	CZS, PW	CZS, PW
V-5.0	A	CZS, PW	CZS, PW	CZS, PW	CZS, PW

A = amorphous, W = wollastonite, CZS = $2\text{CaO}-\text{ZrO}_2-4\text{SiO}_2$, PW = pseudo-wollastonite.

ceramics. In all cases vanadium oxide may be assumed to remain into the residual glassy phase since no crystal containing vanadium was detected.

As confirmed by SEM observation, the crystallisation process for all samples starts from the glass surface and grows monodimensionally inwards the glassy matrix. The complete crystallisation of the B-glass, V-0.1, V-0.3 and V-0.5 samples occurs at 1000 °C, for V-0.7, V-1.0 at 950 °C while V-2.0 and V-5.0 are totally crystallised at

900 and 850 °C respectively. Concerning the samples treated at 1000 °C which contain less than 0.5 mol% of V_2O_5 , the glassy phase appears homogeneous while for the V-0.7 and V-1.0 ones the remaining amorphous phase presents a phase separation (Fig. 11) very similar to what already seen in V-5.0 (Fig. 2). The crystallisation in the V-5.0 sample starts from the surface but with increasing temperature a feeble bulk crystallisation was also shown. SEM observations reveal several characteristic microstructures dependently upon

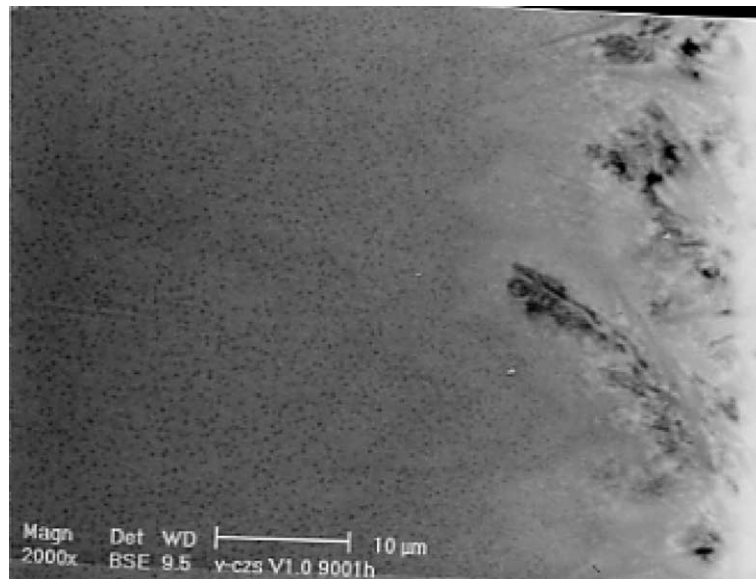


Fig. 11. SEM micrograph of V-1.0 sample heated at 900 °C for 1 h.

V_2O_5 amount so that it is possible to distinguish four classes of materials: B-glass and V0.1 glassy compositions heated at the maximum temperature (Fig. 12) show only dendritic crystalline habitus of

wollastonite, while samples V-0.3 and V-0.5 separate a white crystalline phase rich in ZrO_2 , $2CaO \cdot 4SiO_2 \cdot ZrO_2$, which grows on the outer surface of the wollastonite crystals (Fig. 13). In the

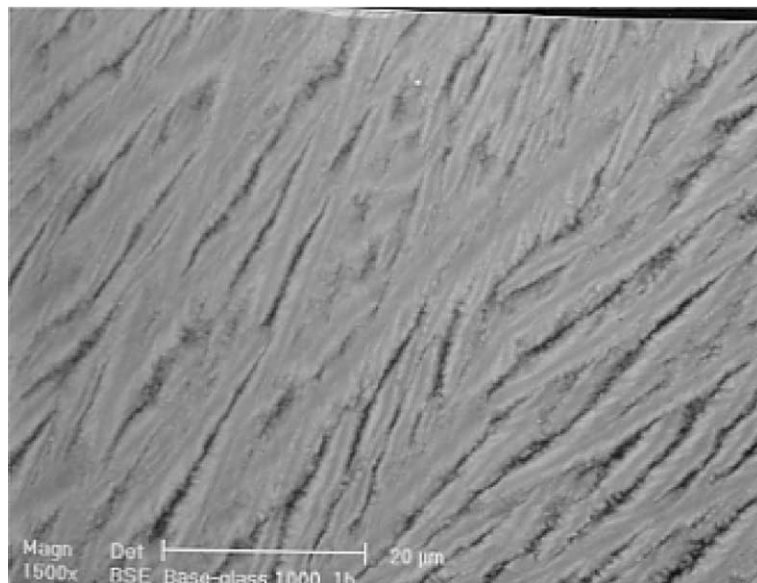


Fig. 12. SEM micrograph of B-glass sample heated at 1000 °C for 1 h.

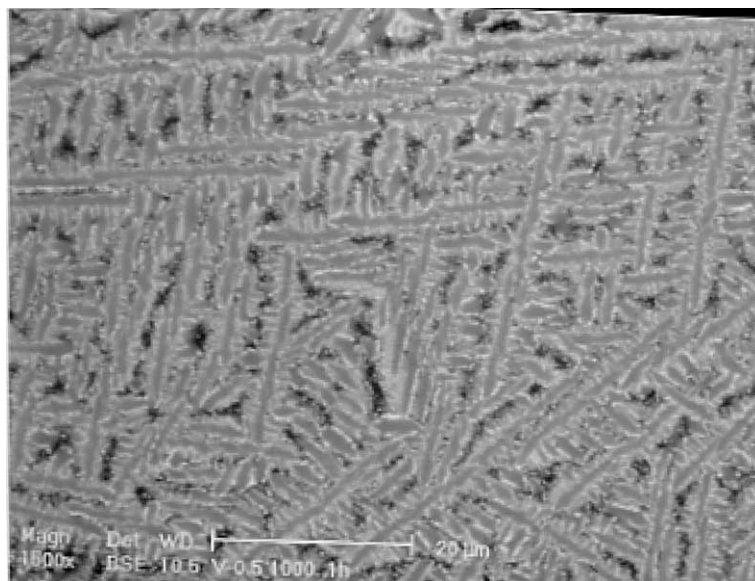


Fig. 13. SEM micrograph of V-0.5 sample heated at 1000 °C for 1 h.

third class oriented long-shaped grey and bright crystals of CaSiO_3 and $2\text{CaO} \cdot 4\text{SiO}_2 \cdot \text{ZrO}_2$ respectively are evident into V-0.7, V-1.0, and V-2.0 (Fig. 14) even if this last composition present also small dark zone. Finally V-5.0 sample presents a

very peculiar microstructure (Fig. 15) with white round shaped crystals ($<5 \mu\text{m}$) of $2\text{CaO} \cdot 4\text{SiO}_2 \cdot \text{ZrO}_2$ and grey round shaped crystals ($<5 \mu\text{m}$) of pseudowollastonite. As confirmed by EDS analysis, the dark areas containing only silicon and

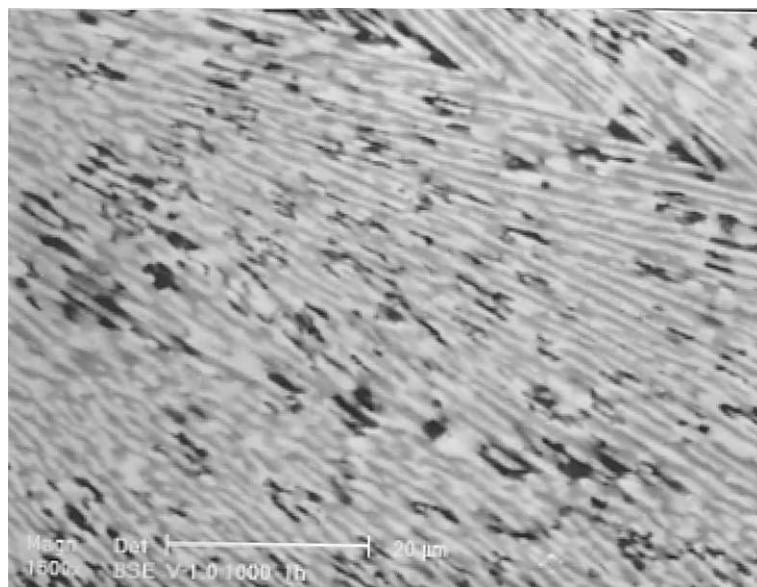


Fig. 14. SEM micrograph of V-1.0 sample heated at 1000 °C for 1 h.

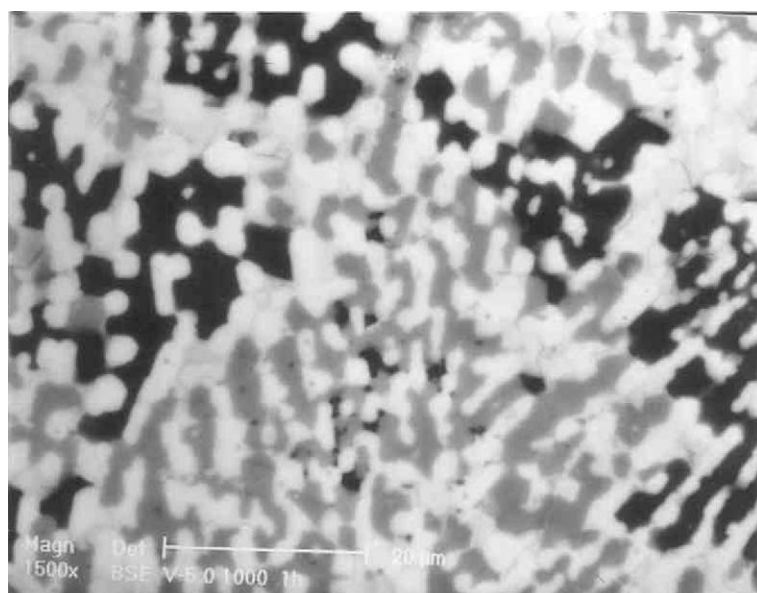


Fig. 15. SEM micrograph of V-5.0 sample heated at 1000 °C for 1 h.

oxygen are randomly dispersed among this crystals network. Up to now, it is not possible to discriminate the nature of these dark zones.

4. Discussion

The appearance of a glass that contains an immiscibility texture depends on the composition and the temperature at which the liquid enters the immiscibility dome on cooling [6]. If the latter occurs at very high temperature, 1500 °C, when diffusion rates are still high, a coarse liquid immiscibility texture results even on rapid cooling. The visual appearance of a glass containing an immiscibility texture depends on the scale of the textures relative to the visible light wavelength. Light scattering due to the presence of these colloidal particles could be the cause of the loss of transparency. So, when the V_2O_5 concentration exceeds 5 mol% a liquid–liquid phase separation during cooling is observed. As a matter of fact V-5.0 sample appears opaque since its texture is on the scale of the visible light wavelengths (0.5–1 μm average diameter), and there is a large difference in refractive index between SiO_2 glass and the residual glassy phase. Liquid–liquid immiscibility is a feature of silica-rich glasses. Immiscibility may be qualitatively regarded as a result of a competition between the modifier cation and silicon for the oxygen atom. Modifier cations prefer a co-ordination environment of non-bridging oxide ions and this tends to break up a melt into silica-rich and silica-deficient regions. In the glasses under study there is the possibility to have high amount of vanadium acting as a glass modifier which induces immiscibility domes.

As reported in Fig. 5, the decrease of T_g as V_2O_5 is added can be explained by taking in account that, according to Ray [7], glass transformation temperature depends on the density of covalent cross-linking and the number and strength of the cross-links between the cation and the oxygen atom. The effect of an element on the glass properties varies with the number of oxygen atoms arranged (co-ordinated) around it. The rated co-ordination number of vanadium, $\Psi = 0.386$, lies at the limit of the tetrahedral arrangement of oxygen

atoms; together with its high-bond strength, this creates conditions for the glass-forming ability of V_2O_5 but at the same time vanadium belongs to those elements with imperfect network-forming ability (group V). This peculiarity, Ψ is responsible of the different behaviours that vanadium oxide can assume into the glassy network. As already reported in a previous work [3] vanadium oxide can exhibit into base-glass three different valences: V^{5+} acts as a former network or as an intermediate; V^{4+} which present octahedral (six co-ordination) or distorted tetrahedral (four co-ordination) sites shows an intermediate behaviour while V^{3+} having octahedral (six co-ordination) site seems to act as a cation modifier. In particular, V^{4+} and V^{3+} are stable at higher amount (up to 1 mol%) of vanadium oxide. Since V_2O_5 decreases T_g temperature especially at higher content of V_2O_5 , the colouring oxide behaves primarily as a network modifier. With this in mind we can also find the explanation for the decreasing trend of the T_p temperature.

At a constant heating rate, ϕ , a broad crystallisation peak indicates a surface crystallisation mechanism, whereas a sharp peak implies a bulk crystallisation process. Consequently, glasses of composition V-1.0 and V-2.0, which show a sharpening of the exothermic peak with respect to the B-glass and low- V_2O_5 content compositions, present a partial bulk crystallisation due to the phase separation formed during heating. The decrease in crystallisation peak height, $(\delta T)_p$ with increasing particle size (Fig. 7) suggests that particles crystallise by a surface nucleation mechanism, particularly evident for the samples containing low- V_2O_5 content. In fact, V-0.1 sample shows a more evident decreasing of $(\delta T)_p$ with respect to V-2.0 sample as particle size increases, indicating a dominant surface nucleation accompanied by a partial internal crystallisation [8,9]. The crystallisation peak temperature, T_p , increases with particle size (Fig. 8). This is due partially to particle size effects on heat transfer. At a given heating rate, larger particles have a greater heat resistance, so it takes longer for centre of the particles to reach the furnace temperature, and a higher observation temperature results [10]. The crystallisation mechanism for all glasses is mainly

a surface mechanism up to 0.7 mol% of V_2O_5 , where the reduction of the extrapolated onset crystallisation temperature is about 50 °C for the finely powdered samples compared to the coarse ones, while for the V-2.0 composition a decrease of about 30 °C is observed.

SEM micrograph of the V-1.0 glass, crystallised at 900 °C for 1 h, is shown in Fig. 11. A crystallised layer growing inwardly from the external surface is clearly visible while the glassy phase presents a phase separation. The microstructure of this glassy phase separation is very similar to the phase separation observed for V-5.0 glassy sample (Fig. 2). Heat treatment on an as-quenched glass is known to promote a coarsening of a liquid–liquid phase separated structure. In the understudied materials this effect is particularly true for the samples containing higher amount of V_2O_5 . Moreover, it is also known the phase separation may promote an internal nucleation mechanism since the separated droplet like phase can acts as nucleation site [11].

As previously reported, the crystallisation activation energy for the different glassy compositions decreases up to V-0.7 composition. The initial decrease in activation energy at lower content of V_2O_5 is due to the fact that the crystal growth activation energy, E , is usually directly dependent to the viscous flow activation energy [12]. As mentioned above vanadium oxide acts as a modifier leading to a more released glass structure and consequently to a lower viscosity which implies an E decrease. On the contrary, an inversion in the E values occurs with increasing the V_2O_5 content up to 2 mol%. Such an increase may have two different reasons: a new crystalline phase is formed in addition to the previous one or a new nucleation mechanism may appear in addition to the already present one. In order to understand the most probable cause, XRD have been collected and analysed. According to XRD results collected at room temperature on the glass-ceramic materials, the first explanation is probable since there two phases have been detected for compositions corresponding to the increase in activation energy. To better follow the crystals' formation in situ high-temperature XRD on glassy powdered samples have been performed. The results (Fig. 16) show that only one phase, wollastonite, appears up to V-

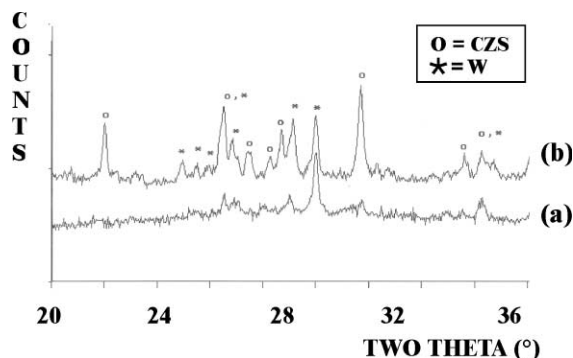


Fig. 16. In situ XRD at 1000 °C for powdered sample: (a) V-0.5 and (b) V-2.0.

0.7 sample while V-1.0 and V-2.0 samples present primarily a new ternary crystalline phase, $2CaO-4SiO_2-ZrO_2$.

The second possible explanation, regarding a different crystallisation mechanism, is sustained by the already reported DTA results and SEM observations: samples containing low amount of V_2O_5 (up to 0.7 mol%) present only surface nucleation mechanism while V-1.0 and V-2.0 samples exhibit also a feeble internal crystallisation.

As previously considered, the studied samples show extremely diverse microstructures. In general, small amounts of vanadium oxide do not strongly influence the B-glass microstructure since there are not relevant differences either in the present crystalline phases either in the microstructure of the doped glasses. Microstructure, which starts to change in the V-0.3 sample, progressively evolves with the increasing of vanadium amount up to 5 mol%, showing remarkable differences imputable to the different crystalline phases formed.

5. Conclusions

Thermal analysis measurements performed at various heating rates on glasses with different amount of V_2O_5 were reported and discussed. T_g and T_p temperatures decrease with increasing V_2O_5 content, confirming the modifier role of this colouring oxide. The DTA peak heights $(\delta T)_p$ decrease with increasing particle size, especially for low- V_2O_5 content, where the crystallisation is primarily a

surface crystallisation mechanism. A feeble internal crystallisation is shown in the V-2.0 sample.

The crystallisation activation energy values show a parabolic trend since E decreases from B-glass (495 ± 7) to V-0.7 (420 ± 6) glasses but with higher amount of V_2O_5 the E values increase up to 442 ± 5 kJ/mol K.

Finally since the glass-ceramic materials obtained from the studied materials present low-crystallisation temperatures and the glass ceramics obtained are greenish and bluish coloured, it is possible to use the studied glasses as coloured frits for tile glazes; moreover the amount of V_2O_5 is almost reduced, the costs of these new materials will be not so different with respect the industrial frits belonging to the ternary base system.

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